

# A Fuzzy AHP-based Quantitative Framework to Prioritize the Crowd-Based Requirements

Saif Ur Rehman Khan<sup>1,\*</sup>, Maha Younus<sup>2</sup>, Javed Iqbal<sup>2</sup>, and Muhammad Abdul Basit Ur Rahim<sup>3</sup>

<sup>1</sup>Department of Computing, Shifa Tameer-e-Millat University, Islamabad, Pakistan

<sup>2</sup>Department of Computer Science, COMSATS University Islamabad (CUI), Islamabad, Pakistan

<sup>3</sup>Computer Engineering and Computer Science, California State University Long Beach, 1250 Bellflower Blvd, Long Beach, 90840, CA

saif\_rehman.ssc@stmu.edu.pk, mahayounuskhan@gmail.com, javediqbal@comsats.edu.pk, m.basit@csulb.edu

\*corresponding author

**Abstract**—Software requirement plays a vital role in the success of a software project. However, the quality of requirement management remains a key challenge for a project manager, especially in a Crowdsourcing context. In Crowdsourcing, the customers belong from diverse locations; thereby, providing the redundant and inconsistent requirements for a targeted software system. To the best of our knowledge, the existing literature lacks in providing a quantitative framework that prioritizes the crucial factors affecting the requirement's management in the Crowdsourcing context. Motivated by this, we propose a quantitative framework that will be effective in ranking the Crowdsourcing-based requirements. For this purpose, the current study intends to identify various factors, that can affect the management of requirements, from the literature and also validates them from the practitioners. To accomplish this, we perform a systematic review and an empirical study. Moreover, we quantify the impact of each factor using the capability of a multi-criteria decision-making oriented Fuzzy- Analytical Hierarchy Process (F-AHP) technique. Subsequently, we also employ a multi-perspective prioritization technique to obtain the data from the relevant personnel to validate the quantified score. To assess the performance of the proposed quantitative framework, we considered a real-world Smart Home case study. Moreover, the efficacy of the adopted F-AHP technique is validated by comparing it with the traditional AHP technique. The results indicate that the F-AHP technique significantly outperformed the baseline technique.

**Keywords**—crowd-based requirements engineering; quantitative framework; multi-criteria decision-making; fuzzy AHP; systematic review; empirical study.

## 1. INTRODUCTION

Crowdsourcing [1] is a collection of information ranging from market surveys to obtaining the general opinion of the crowd. Crowdsourcing has been applied in all research domains including Requirements Engineering (RE); thus, coining the notion of Crowd-Based Requirements Engineering (CBRE). Traditionally, RE is a pivotal part of the software development life cycle as it is crucial for understanding and managing stakeholders'

needs [2]. It covers all the activities that are involved in discovering, documenting, and maintaining the requirements of a product. The software requirements are usually gathered from a set of stakeholders or user representatives, which are further specified by requirements engineers [3]. Furthermore, the requirements are processed from raw requirements to structured requirements via RE activities such as analysis, modeling, and management. Therefore, effective RE is regarded as the most basic and vital building block for the development of a software product. The CBRE is an enveloping term, for approaches either automated or semi-automated, for gathering or analyzing data from crowds to obtain user requirements [4]. It allows to obtain an understanding of the users' wants and needs for a product. CBRE intends to gather the requirements from a large group of individuals of varying knowledge, personalities, identities, and nationalities. It integrates existing elicitation techniques with a crowd to fulfill the existing limitations in the RE process [5].

The developers can potentially access a wider range of users' perceptions and gain knowledge essential for meeting their needs and expectations. From a future implication viewpoint, Crowdsourcing provides insight into future trends through the users' perception. Hence, it reduces the cost as well as providing access to a wide range of public opinions that lead to a potentially successful product. However, Crowd RE faces several challenges, such as quality assurance, since the requirements are primarily obtained from the novice crowd. Hence, the quality of requirements can be of varying degrees. Moreover, it can give rise to legal issues, and business viability [6][7][8]. Human error is regarded as a crucial factor in Crowd RE in which the users might provide ambiguous requirements due to lack of knowledge. Moreover, crowdsourcing includes other issues such as overlooking minorities and malicious participants, as well as unhealthy dependencies among the crowd. To apply crowdsourcing on a large scale, software companies require a considerable level of openness, i.e., users should be able to view the ideas of the organization and other users.

Crowdsourcing in the RE context is gaining the wide attention of researchers. At present, CBRE has primarily focused on the acquisition of requirements by using ML techniques such as NLP [9][10] and data mining [11][12] to extract requirements from platforms including App

reviews. Moreover, discussion boards [13] are created to provide a platform for crowds to interact and provide requirements. Furthermore, Gamification [14][15] has also been applied primarily to motivate the crowd to participate. However, one of the main challenges while employing Crowd RE is the quality and quantity of requirements [13]. Due to the involvement of a crowd of varying numbers, the obtained requirements can be plentiful, and their quality might be compromised. Thereby, it highlights the need to organize the obtained requirements in a well-structured format. With the tremendous advancement and research in Crowd RE, it is surprising to notice that there is a lack of prioritization processes for the obtained requirements.

Motivated by this, we propose a quantitative framework for prioritization of the Crowd-based requirements. For the data collection phase, we identify the factors that affect the requirements prioritization process by conducting a systematic literature review. Next, we validate the level of impact of each criterion from the practitioners through a questionnaire-based survey. After calculating the level of impact, we obtain the data values of each of the requirements against each criterion from the relevant personnel. Finally, the proposed framework prioritizes the Crowd-based requirements by adopting a Multi-Criteria Decision-Making (MCDM) oriented F-AHP technique, which considers the identified criteria and their decision weights. In summary, the following are the main Research Contributions (RCs) of this work:

- RC1: Identify the influencing factors through a SLR protocol in the CBRE context.
- RC2: Determine the impact of each criterion on the requirements prioritization process using a questionnaire-based survey by involving the practitioners.
- RC3: Application of the proposed quantitative framework using a real-world case study.
- RC4: Validation of the proposed framework by comparing results of the proposed F-AHP technique with traditional AHP.

The rest of the paper is organized as follows: Section 2 discusses the related work, while the adopted research design is mentioned in Section 3. Section 4 presents a detailed description of the proposed quantitative framework. The results are reported and analyzed in Section 5. Section 6 discusses the findings of the study. Finally, Section 7 concludes the current work, and also outlines potential future research dimensions.

## 2. RELATED WORK

This section provides the related work about the CBRE context. Saeed et al. [8] proposed a quantitative framework to prioritize the identified cost factors grounded on the 4Ps (i.e., People, Project, Process, and Product) standard in the

agile software development context. Conversely, Peng et al. [16] proposed a VOC (voice of customer) framework for business prioritization by using the crowd. The authors analyzed the VOC data and extracted useful knowledge by integrating domain knowledge and data-driven approaches to obtain functional requests. However, with the increase in customer requests, the proposed framework lacks in effectively handling the customer's requests. In contrast, Musto et al. [17] proposed a domain-dependent framework (i.e., CrowdPulse) to analyze text stats in social streams. It extracts data from social networks and implements algorithms for semantic processing, sentiment analysis, and opinion mining to classify them into positive and negative attributes. Conversely, Makrynioti et al. [18] reported on a business intelligence platform that used crowd knowledge expressed in social media. Moreover, the authors filtered the data in real time as well as summarized the historical content by using data mining techniques.

In contrast, Kolpondinos et al. [19] proposed a social media platform (i.e., GARUSO) to solve the problem of motivating the stakeholders through a gamification technique. However, they reported the bias in the attained results due to the voluntary participation of the stakeholders. From the multi-criteria prioritization facet, Ibrishet et al. [20] proposed an algorithm by considering a Multiple Perspective Prioritization Technique (MPPT) that represents the business, technical, and customer perspectives for a medium to a large number of requirements. The authors reported that the MPPT technique yielded better results than the considered prioritization techniques.

In contrast, Dabbagh et al. [21] aimed to reduce the subjectivity in the results by applying an MCDM for requirements prioritization and conducted a comparative analysis on three prioritization approaches namely IPA, AHP, and HAM to prioritize both functional and non-functional requirements. Conversely, Mougouei et al. [22] proposed a framework for partial selection and prioritization of security requirements using fuzzification. The authors mentioned that their proposed framework prioritized the security requirements based on factors such as cost, technical complexity, and impact. In comparison to the discussed works, Khan et al. [23] applied F-AHP to prioritize the success factors in the context of Global Software Development (GSD). They ranked the success factors and created a thematic taxonomy of factors based on the results of the conducted questionnaire-based survey.

In summary, the current state-of-the-art lacks in identifying, validating, and prioritizing the influential factors of CBRE using the F-AHP technique. Moreover, there is a need to develop a quantitative framework that supports the above-mentioned limitations in the context of CBRE.

### 3. RESEARCH DESIGN

This section describes the adopted research methodology containing three main phases. Firstly, a systematic review was conducted to identify the key factors influencing the requirements prioritization process. Next, a questionnaire-based survey was performed to obtain the user requirements data from the practitioners. Finally, an F-AHP-based approach was adopted to prioritize the requirements grounded on the identified influential factors.

#### 3.1. Systematic Literature Review (SLR)

The purpose of conducting an SLR is to identify the influential factors/criteria affecting the prioritization process in CBRE by finding the relevant primary studies in an unbiased and systematic manner. To accomplish this, we followed the recommendations of Kitchenham et al. [24] and adopted three main phases of SLR: (i) review planning, (ii) review conducting, and (iii) review reporting.

##### 3.1.1. Review Planning

In the review planning phase, we performed the suggested steps discussed in the following sections:

###### 3.1.1.1. Research Question

*RQ1: What are the key influential factors impacting requirement prioritization in the CBRE context?*

Rationale: RQ1 aims to find the influential factors useful for ranking the requirements in CBRE. Ultimately, it supports devising a requirements prioritization mechanism. The following sections must be numbered according to the format

###### 3.1.1.2. Database Selection

To find the most relevant primary studies, we considered the widely employed electronic databases including ACM, IEEE Xplore, Science Direct, and Wiley Online.

###### 3.1.1.3. Search String

To find the relevant primary studies, we devised a search string grounded on the RQ1. For this purpose, we identified the main keywords and their synonymous and joined them using the logical connectives, i.e., 'AND', 'OR'. For example, the devised search string uses the keywords "crowd", "requirements prioritization" and "factor" and connects the main keywords using the logical connector 'AND' to find the potential factors from the current state-of-the-art CBRE domain.

((("Crowd" OR "Crowd-Based") AND ("Requirement Prioritization" OR "Requirement Ranking") AND ("Factor" OR "Criteria" OR "Influential Factor"))

###### 3.1.1.4. Inclusion and Exclusion Criteria

The search string was applied to the selected electronic databases. However, many irrelevant studies were included in the results due to generic search keywords in

the defined search string. Therefore, we selected the potentially relevant studies according to pre-defined inclusion and exclusion criteria.

The Inclusion Criteria (IC) for the selection of studies are as follows:

- IC1: The potential study must be written in the English Language.
- IC2: The potential study must be published in a peer-reviewed journal, conference, or workshop proceeding.
- IC3: The potential study must focus on the CBRE context.
- IC4: The publication must range from the year 2010 to the year 2024.

The following Exclusion Criteria (EC) was applied:

- EC1: The studies focusing other than the CBRE context were excluded.
- EC2: The grey literature including lecture slides, tutorials, editorials, and blogs were excluded due to insufficient technical details.
- EC3: The short studies (containing only 3 to 4 pages) were excluded.
- EC4: The studies with short or graphical abstracts were excluded due to insufficient technical details.

###### 3.1.1.5. Selection of Studies

The potential studies were selected based on the defined inclusion and exclusion criteria. Table 1 shows the distribution of studies selected from the considered electronic database. The duplicate studies were manually removed by checking their title. In the SLR, we used a tollgate approach to refine the selected studies.

The tollgate approach includes the following steps:

- Step 1: Use search strings and inclusion criteria to search the potential studies from the considered electronic databases.
- Step 2: The studies were included or excluded by checking their title and abstract.
- Step 3: The studies were included or excluded by checking the introduction and conclusion sections.
- Step 4: The studies were included or excluded based on their full text.
- Step 5: Select the final studies based on the result of defined quality assessment criteria.

Initially, 106 potential studies were selected by applying the devised search string on the selected electronic databases. Finally, 17 primary studies were selected by applying the tollgate approach. Table 1 describes the statistics of the selected studies from the selected electronic repositories.



Table 1. The statistics about the potential studies

| Database       | Potential studies | Excluded | Included |
|----------------|-------------------|----------|----------|
| IEEE           | 43                | 41       | 2        |
| Science Direct | 13                | 6        | 7        |
| Wiley          | 19                | 14       | 5        |
| ACM            | 31                | 28       | 3        |
| Total          | 106               | 89       | 17       |

To answer the devised RQ1, we have extracted the influential factors for effective requirement prioritization in the CBRE domain. According to the literature, several factors can negatively impact the prioritization process. In total, 12 influential factors were identified from the literature as shown in Table 2.

Table 2. Influential factors affecting requirement prioritization in CBRE context

| Sr# | Factors                     | References                                                   | Frequency |
|-----|-----------------------------|--------------------------------------------------------------|-----------|
| 1   | Time/ Effort                | [25][26][27][28][29][30][31][32][20][33][34][35][36][37][38] | 88%       |
| 2   | Cost/ Budget                | [20][25][26][27][29][30][31][32][33][34][35][37][39][40]     | 82%       |
| 3   | Resource/Skill Availability | [26][29][30][31][32][35][39][41][42]                         | 53%       |
| 4   | Market Need                 | [30][32][39][40][43]                                         | 29%       |
| 5   | User Preference             | [36][37][40][42][44]                                         | 29%       |
| 6   | Geographical Area           | [30]                                                         | 5.9%      |
| 7   | Analyst Qualification       | [30]                                                         | 5.9%      |
| 8   | Dependencies                | [26][31][32][34][35][38]                                     | 35%       |
| 9   | Biasness                    | [26][45][46][47]                                             | 24%       |
| 10  | Risk                        | [20][38]                                                     | 12%       |
| 11  | Implementation Difficulty   | [20]                                                         | 5.9%      |
| 12  | Scalability                 | [38][48]                                                     | 12%       |

From Table 2, it is evident that time (88%; 15/17) and cost (82%; 14/17) are the most frequently reported factors in the literature. Thus, it can be concluded that time and cost factors can play a crucial role in the decision-making process. In contrast, the skills (53%; 9/17) and preferences (29%; 5/17) were reported in the selected studies. Furthermore, it can be observed from Table 2 that some factors have a higher impact on the requirements compared to the other factors found from the literature. For example, the availability of the resource (53%; 9/17) and the user preferences (29%; 5/17) are also important as the product is for the user and it can only be created if the appropriate resource or skill is available. Conversely, other factors such as market need (29%; 5/17), geographical area (5.9%; 1/17), the qualifications of the analyst (5.9%), dependencies among requirements (35%; 6/17), potential risk (12%; 2/17), difficulty in implementation (5.9%; 1/17), and scalability (12%; 2/17) have also been discussed in the literature (Table 2). As the difficulty level increases, the priority level of the requirement diminishes to some

extent. Notice that every factor is directly proportional to the priority of the requirement such as preference, skill, need, and so on. Alternatively, it can be inversely proportional, i.e., an increase in the factor value decreases the priority level and vice versa.

#### 4. PROPOSED FRAMEWORK

This section presents the proposed framework for ranking of the requirements in the CBRE domain. Figure 1 represents the proposed framework. The proposed framework contains two main phases. In the first phase (i.e., data collection), we identified the factors influencing the CBRE through the performed SLR (as reported in Section 3). While, the second phase (i.e., data processing), presents the proposed framework intended to prioritize the crowd-based requirements.

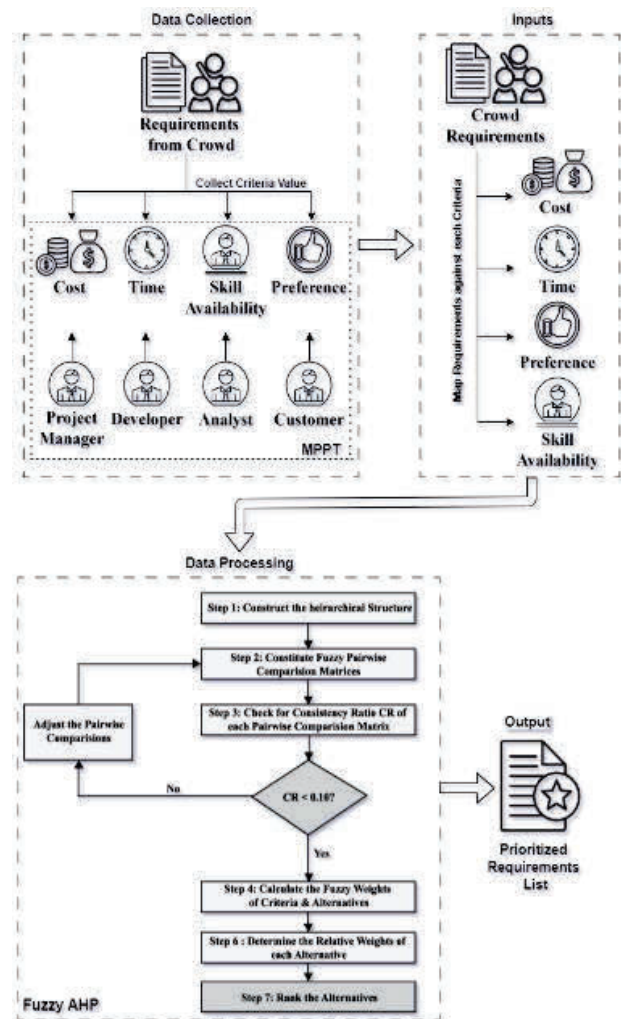


Figure 1. The proposed quantitative framework

Notice that the proposed framework can be viewed as a hybrid model that applies two techniques for the

prioritization of Crowd-based requirements. The first technique is the MPPT, which permits to gain the crowd's insights for each influential factor/criterion. The second technique is an F-AHP. We used the values obtained through MPPT and applied the F-AHP technique to prioritize the Crowd-based requirements according to the identified criteria (Figure 1).

In this work, we identified four factors/criteria (i.e., cost, time, preference, and skill availability) that influence the Crowd-based requirements. As previously mentioned, the factors/criteria were identified through a SLR and selected based on their frequency analysis. The selected factors/criteria were mapped against a set of Crowd-based requirements and finally, the requirements were prioritized by following four-factor prioritization using F-AHP technique. To reduce the subjectivity or biases in the prioritization process, four different people with different roles were involved in the data collection process to obtain the values of each requirement against each factor/criterion individually. This is due to the reason that the main stakeholders in this process are the crowd; hence, the crowd preference holds the highest impact on the requirements prioritization process.

The second most important criterion was skill availability. Cost and time hold equal importance in the prioritization process. Hence, these four factors were used as criteria for the employed MCDM technique. The next step was the determination of a data set and obtaining the values of the requirements against each criterion/factor. Each of the criteria values was obtained from relative personnel who have the highest knowledge regarding that factor. The preference of each requirement was obtained from the crowd from a customer perspective. Similarly, the time taken for its accomplishment was obtained from the developer. Likewise, a project manager provided the estimated costs, and a requirement analyst provided the ease of resource availability.

By following the MPPT process, we obtained the perspective of multiple crowds (roles), it reduces the subjectivity of the results. The next step of the research is the application of the F-AHP. First the construction of a hierarchical structure was done, where the main goal and the factors, and alternatives were identified to justify the main goal (Figure 2).

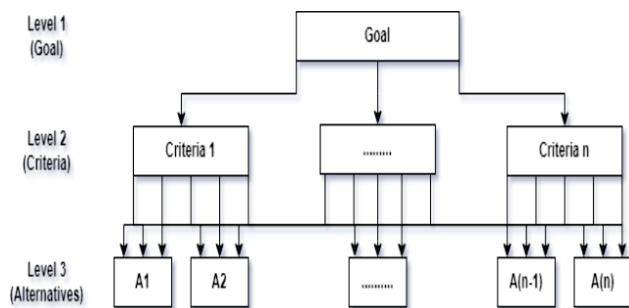


Figure 2. Hierarchical structure

The next and most crucial part of the process is to construct the pairwise comparison by calculating the fuzzy weights of the criteria. After assigning weights to the criteria, the next step is to assign weights to each alternative (i.e., the requirements against each criterion). After the assignment of weights to the criteria and the alternatives, it is required to compute the relative weight of each alternative and rank them accordingly.

#### 4.1. Fuzzy Analytical Hierarchy Process

This section illustrates the underlying working mechanism of an F-AHP technique.

##### 4.1.1. Fuzzy Set Theory

Fuzzy set theory was proposed by Zadeh [49] in 1965 as a generalization of conventional set theory and was applied to problems under uncertain circumstances. When dealing with imprecise knowledge and ambiguous information, fuzzy sets are applied to reduce personal bias. It is characterized by a membership function that only has objects between '0' and '1'. The traditional set theory was developed to deal with Boolean logic problems. However, the idea behind fuzzy sets is to represent the degree to which elements belong to the specific sets. In conventional set theory, a characteristic function is used as a mapping function. However, in fuzzy set theory, a membership function is used to define a fuzzy set.

A fuzzy number is a fuzzy set in which the membership function satisfies the condition of normality and convexity, as formally defined in Equation 1 [49], as follows:

$$\text{SUP}_{x \in X} \mu_A(x) = 1 \dots \dots \dots (1)$$

where SUP means the "least upper bound" value.

##### 4.1.2. Triangular Fuzzy Number

Triangular Fuzzy Number (TFN)  $\tilde{T}$  is denoted by  $(t^l, t^m, t^u)$  and mathematically defined in Equation 2 [49], as follows:

$$\mu_T(x) = \begin{cases} \frac{x-t^l}{t^m-t^l}, & t^l \leq x \leq t^m \\ \frac{(t^u-x)}{t^u-t^m}, & t^m \leq x \leq t^u \\ 0 & \text{Otherwise} \end{cases} \dots \dots (2)$$

For any two fuzzy numbers  $\tilde{T}_1 = (t_1^l, t_1^m, t_1^u)$  and  $\tilde{T}_2 = (t_2^l, t_2^m, t_2^u)$ , the operational laws of triangular fuzzy numbers are described in Table 2.

By using the same Fuzzy numbers mentioned in Table 3, the distance between  $\tilde{T}_1$  and  $\tilde{T}_2$  can be calculated using Equation 3 [49]:

$$\text{Distance}(\tilde{T}_1, \tilde{T}_2) = \sqrt{\frac{1}{3} [(t_1^l - t_2^l)^2 + (t_1^m - t_2^m)^2 + (t_1^u - t_2^u)^2]} \dots \dots \dots (3)$$

Table 3. Triangle fuzzy number operational law

| Operational Law                                      | Expression                                                                                                           |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Addition $(\widetilde{T}_1 + \widetilde{T}_2)$       | $(t_1^l, t_1^m, t_1^u) + (t_2^l, t_2^m, t_2^u)$<br>$= (t_1^l + t_2^l, t_1^m + t_2^m, t_1^u + t_2^u)$                 |
| Subtraction $(\widetilde{T}_1 - \widetilde{T}_2)$    | $(t_1^l, t_1^m, t_1^u) - (t_2^l, t_2^m, t_2^u)$<br>$= (t_1^l - t_2^l, t_1^m - t_2^m, t_1^u - t_2^u)$                 |
| Multiplication $(\widetilde{T}_1 * \widetilde{T}_2)$ | $(t_1^l, t_1^m, t_1^u) * (t_2^l, t_2^m, t_2^u) = (t_1^l * t_2^l, t_1^m * t_2^m, t_1^u * t_2^u)$                      |
| Division $(\widetilde{T}_1 \div \widetilde{T}_2)$    | $(t_1^l, t_1^m, t_1^u) \div (t_2^l, t_2^m, t_2^u) = (\frac{t_1^l}{t_2^u}, \frac{t_1^m}{t_2^m}, \frac{t_1^u}{t_2^l})$ |
| Inverse $(\widetilde{T}_1)^{-1}$                     | $(\widetilde{T}_1)^{-1} = (\frac{1}{t_1^u}, \frac{1}{t_1^m}, \frac{1}{t_1^l})$                                       |
| For any Real Number $k$                              | $k(t_1^l, t_1^m, t_1^u) = (kt_1^l, kt_1^m, kt_1^u)$                                                                  |

To understand the F-AHP process, it is essential to first understand the underlying workings of the AHP technique. The following are the preliminary steps of AHP:

- Step 1: Construction of hierarchical structures of the targeted problem.
- Step 2: Computation of the criteria weight by applying pairwise comparisons at each hierarchy level.
- Step 3: Calculate the weights of alternatives using a pairwise comparison matrix.

The above-mentioned are the basic steps of the AHP process. In contrast, F-AHP [22] is the integration of fuzzy set theory with the AHP technique. Although, both are MCDM techniques; however, the F-AHP technique is applied in uncertain circumstances or ambiguous data due to its capability of providing more accurate and adequate judgments.

#### 4.1.3. Linguistic Variables

Linguistic variables are words in natural language that define the membership functions. Each membership function has three parameters which are the triangular fuzzy numbers that identify the ranges of the function [49]. Table 4 shows the linguistic values and their corresponding fuzzy values.

The level of importance of each alternative can be determined by using the scale mentioned in Table 4. The linguistic variables are used for the user's ease in obtaining values for each alternative, accordingly.

Table 4. Linguistic variable fuzzy scale

| Linguistic Variable                  | AHP Scale | Fuzzy Triangular Scale |
|--------------------------------------|-----------|------------------------|
| Equally Important                    | 1         | (1,1,1)                |
| Equally to Moderately Important      | 2         | (1,2,3)                |
| Moderately Important                 | 3         | (2,3,4)                |
| Moderately to Strongly Important     | 4         | (3,4,5)                |
| Strongly Important                   | 5         | (4,5,6)                |
| Strongly to Very Strongly Important  | 6         | (5,6,7)                |
| Very Strongly Important              | 7         | (6,7,8)                |
| Very Strongly to Extremely Important | 8         | (7,8,9)                |
| Extremely Strongly Important         | 9         | (8,9,9)                |

#### 4.2. The Fuzzy-AHP Technique

In this section, a step-by-step process of the F-AHP technique is discussed [49].

**Step 1:** The first and most vital step of the F-AHP is the construction of a pairwise comparison matrix of all the criteria of the hierarchy. Equation 4 [49] shows the pairwise comparison matrix  $\widetilde{T}^k$  with members  $t_{ij}^k$ :

$$\widetilde{T}^k = \begin{bmatrix} \widetilde{t}_{11}^k & t_{12}^k & \cdots & \widetilde{t}_{1n}^k \\ \widetilde{t}_{21}^k & \cdots & \cdots & \widetilde{t}_{2n}^k \\ \vdots & \cdots & \cdots & \vdots \\ \widetilde{t}_{n1}^k & t_{n2}^k & \cdots & t_{nn}^k \end{bmatrix} \dots \dots \dots (4)$$

where  $i, j = 1, 2, 3 \dots n$  and  $k$  represent the  $k^{\text{th}}$  decision makers' preference of criteria  $i$  over  $j$  with the help of triangular fuzzy numbers.

**Step 2:** In the case of multiple decision-makers, the preferences of each decision-maker are summed up and their average value is considered to accommodate all decision-maker's preferences and reduce the biases. This process is formally modeled using Equation 5 [49]:

$$\widetilde{t}_{ij} = \frac{\sum_{k=1}^K \tau_{ij}^k}{K} \dots \dots \dots (5)$$

**Step 3:** After taking the average of all decision-maker preferences, the next step is to create a new pairwise comparison matrix or update the existing matrix for all the criteria and alternatives using the averaged preferences of the decision-makers. Equation 6 [49] formally describes the pairwise comparison matrix:

$$\widetilde{T} = \begin{bmatrix} \widetilde{t}_{11} & \cdots & \widetilde{t}_{1n} \\ \vdots & \ddots & \vdots \\ \widetilde{t}_{n1} & \cdots & \widetilde{t}_{nn} \end{bmatrix} \dots \dots \dots (6)$$

**Step 4:** For the calculation of the relative fuzzy weights of each criterion, the geometric mean technique is utilized. Grounded on this technique, the geometric mean is calculated, and fuzzy weights of each criterion are obtained using Equation 7 [49], formally defined as follows:

$$\widetilde{r}_i = \left( \prod_{j=1}^n \widetilde{t}_{ij} \right)^{1/n}, i = 1, 2, 3, \dots, n \dots \dots \dots (7)$$

Notice that each iteration in the pairwise-comparison matrix is a fuzzy number. Thereby, it has three values in each iteration, the lower, middle, and upper bound of the fuzzy triangular number. By taking its reciprocal, the reciprocal of

the upper bound value becomes the lower bound, and the middle bound's reciprocal stays in the middle position as formally defined in Equation 8 [49]. Notice that the reciprocal of the lower bound value in the fuzzy triangular number becomes the upper bound in the reciprocal form.

$$t_{ij} = \frac{1}{t_{ji}} \Rightarrow \begin{cases} t_{ij} = (l_{ij}, m_{ij}, u_{ij}) \\ \frac{1}{t_{ij}} = \left( \frac{1}{u_{ij}}, \frac{1}{m_{ij}}, \frac{1}{l_{ij}} \right) \end{cases} \quad \dots (8)$$

While calculating the product of the triangular fuzzy numbers in the pair-wise comparison matrix, the lower bound is multiplied by the lower bound; the middle bound is multiplied by the middle bound; similarly, the upper bound is multiplied by the upper bound. This process is formally modeled using Equation 9 [49], as follows:

$$\begin{aligned} \tilde{t}_i &= (l_i, m_i, u_i) \\ l_{ij} &= (l_{i1} * l_{i2} * \dots * l_{in})^{1/n} \quad i = 1, 2, 3, \dots, n \\ m_{ij} &= (m_{i1} * m_{i2} * \dots * m_{in})^{1/n} \quad i = 1, 2, 3, \dots, n \\ u_{ij} &= (u_{i1} * u_{i2} * \dots * u_{in})^{1/n} \quad i = 1, 2, 3, \dots, n \end{aligned} \quad \dots (9)$$

**Step 5:** The computation of fuzzy weights criteria is formally defined using Equation 10 [49], as follows:

$$\tilde{w}_i = \tilde{r}_i * (\tilde{r}_1 + \tilde{r}_2 + \dots + \tilde{r}_n)^{-1} \quad \dots (10)$$

**Step 6:** After the calculation of the fuzzy weights, the next step is to normalize the weights. For that purpose, the first step is to compute the average weight of the criteria, which is calculated using Equation 11 [49], mathematically defined as follows:

$$M_i = \frac{\tilde{w}_1 + \tilde{w}_2 + \dots + \tilde{w}_n}{n} \quad (11)$$

**Step 7:** After obtaining the average weights, we normalized the weight of the criteria using Equation 12 [49]. Finally, we arranged them into an ascending order to generate the prioritized results.

$$N_i = \frac{M_i}{M_1 + M_2 + \dots + M_n} \quad (12)$$

## 5. RESULTS AND ANALYSIS

This work reports the results of the conducted survey and analyses them to identify the level of impact of identified factors on the requirements prioritization process. In this work, the data is collected with the help of a questionnaire-based study. For accurate results, the identification notion of the targeted population is essential. Notice that the population is defined as the set of components utilized to acquire the information necessary for examining the deduction made by the participating researcher. Data collection from all members of the targeted population remains a challenging task. For this reason, a set of participants are selected from the concerned

population. In this work, the targeted population is the experts in the Requirement Engineering domain. The targeted audience varies from requirements analysts to project managers, and developers to customers. Data were collected from organizations having 2-5 years or more experience. Surely, the experience and knowledge of the crowd (respondents) about the SE paradigm is essential to attain more accurate results.

### 5.1. Demographic Data of Respondents

Through demographic profiling, we analyses more details about the respondents. The location, experience, and organization size were observed with the help of demographic data. The questionnaires were distributed among 50 participants, which had varying demographic data. Figure 3 shows the demographics of the global participants. It was observed that 33 (out of 50) participants belonged to Pakistan, whereas 8 were from India, and 4, 3, and 1 were from Australia, the USA, and the UK, respectively.

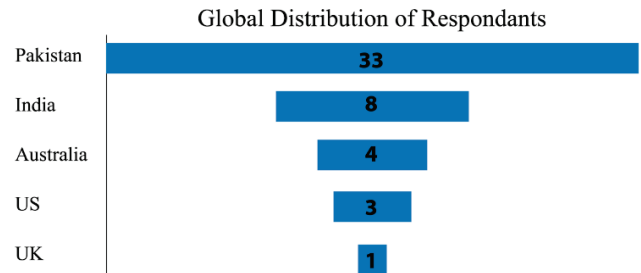


Figure 3. Demographic information of the participants

The respondents have varying levels of experience. For example, 18 respondents had the experience of five or more years, whereas 22 respondents had 3- 5 years of experience, and 10 participants had less than 3 years of experience.

Figure 4 shows the size of the organizations of the respondents/participants. As shown in Figure 4, 46% of the respondents are from a medium-sized organization, 22% are from a small organization, and 32% of the participants are from relatively large organizations. The demographic data concludes that the data was collected from personnel with varying work histories and localities, which is useful in reducing the biases from the obtained results.

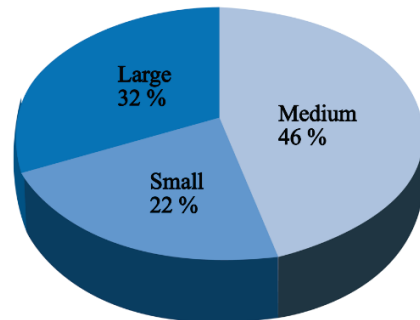


Figure 4. Organizational size of the participants



## 5.2. Response against Factors

The questionnaire was distributed amongst experts in the software development environment involving crowds. The factors identified from the SLR were compared against one another to obtain each factor's level of impact on the crowd-based requirement prioritization process. The results indicated that the majority of the participants agreed that the "preference" had the highest impact on the requirements prioritization process. Budget and time play equal roles in the requirements prioritization process and lastly, skill/resource required. Then, another set of questionnaires was distributed where each of the requirements was mapped against each of the four factors to obtain the individual values of each alternative against each factor/criterion.

### 5.2.1. Application of F-AHP for Prioritizing Crowd-Based Requirements

To prioritize the crowd-based requirements according to the four identified factors, the first step is to construct a pairwise comparison matrix of the factors. The next step is to construct the pairwise comparison matrix of the alternatives against each criterion individually. After that F-AHP technique was applied to a data set of crowd requirements obtained for the construction of a smart home case study. The implementation of the technique was done on a Microsoft Excel 365 environment. The step-by-step implementation of the F-AHP method is explained in the following sections:

#### 5.2.1.1. Step-1 (Hierarchy Structure of the Problem and its Solution)

In the AHP technique, the first step is the construction of a hierarchical structure. The hierarchical structure consists of three levels. Level 1 identifies the main goal of performing AHP, in this case the main goal is to prioritize the crowd-based requirements. In contrast, level 2 explores the criteria/factors to achieve the goal. We found four factors, including cost, time, skill, and preference, through the SLR, and also their level of impact has been identified via a questionnaire-based survey distributed amongst software professionals and project managers. Finally, level 3 of the hierarchical structure discovers the alternatives, (i.e., crowd-based requirements) obtained from the crowd for the construction of a smart home. The hierarchical structure of the targeted problem is shown in Figure 5.

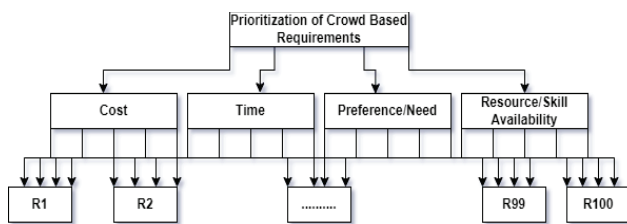


Figure 5. Hierarchical structure for prioritization of crowd-based requirements

#### 5.2.1.2. Step-2 (Conducting the pair-wise comparison)

The main objective of this work is to prioritize the crowd-based requirements grounded on the identified criteria. The

most vital part of the proposed technique is the construction of a pairwise-comparison matrix. For this purpose, we developed four questionnaires and distributed them among 4 different personnel to obtain data values of the alternatives (i.e., the crowd-based requirements). Each questionnaire obtained the values of the requirements against one single criterion.

To perform multi-criteria and multi-perspective decision-making, the data is collected from different perspective personnel and against different criteria to obtain subjective results and reduce biases. The respondents had different designations (e.g., requirement engineer, project manager, developer). Similarly, the time values of each alternative were obtained from a developer's perspective. Likewise, the ease of availability of the skill was obtained from market analysts as they were aware of the norms in the industry. Furthermore, the requirements preference values were obtained from the potential customers. The relatively small sample size of the survey is mainly because F-AHP applies to a smaller data set. The result of a pairwise-comparison matrix of the identified criteria is shown in Table 5, and computed using Equation 13 [49], formally defined as follows:

$$\bar{r}_i = \left\| \left| \bar{t}_n \right| \right\|, i = 1, 2, 3, \dots, n \quad (13)$$

Table 5 shows the priority values of the identified factors. Evidently, the priority value of *preference* is the highest and the *skill availability* has the lowest priority level.

Table 5. Pairwise comparison of identified criteria/factors

|            | Cost            | Time            | Preference | Resource      | Ni    |
|------------|-----------------|-----------------|------------|---------------|-------|
| Cost       | (1, 1, 1)       | (1, 1, 1)       | (8, 9, 9)  | (3, 4, 5)     | 0.428 |
| Time       | (1, 1, 1)       | (1, 1, 1)       | (5, 6, 7)  | (4, 5, 6)     | 0.413 |
| Preference | (19, 19, 1/8)   | (1/7, 1/6, 1/5) | (1, 1, 1)  | (1/3, 1/2, 1) | 0.058 |
| Resource   | (1/5, 1/4, 1/3) | (1/6, 1/5, 1/4) | (1, 2, 3)  | (1, 1, 1)     | 0.101 |

#### 5.2.1.3. Step-3 (Geometric Mean of Fuzzy Comparison Value)

Using Equation 7, we have calculated the relative fuzzy weights of each of the criteria. We applied the geometric mean method and calculated the fuzzy weights of the identified criteria and their role in the requirements prioritization process of the crowd-based requirements. Table 6 shows the fuzzy weights of the criteria. Each factor has three values, including



upper, middle, and lower bounds, which represent the fuzzy weights of each criterion.

Table 6. The geometric mean of fuzzy comparison value

| Criteria        | Ri    |       |       |
|-----------------|-------|-------|-------|
| Cost            | 2.213 | 2.449 | 2.590 |
| Time            | 2.115 | 2.340 | 2.546 |
| Preference      | 0.270 | 0.310 | 0.398 |
| Resource        | 0.427 | 0.562 | 0.707 |
| Total           | 5.025 | 5.662 | 6.240 |
| Reverse         | 0.199 | 0.177 | 0.160 |
| Ascending Order | 0.160 | 0.177 | 0.199 |

#### 5.2.1.4. Step-4 (Geometric Mean of Fuzzy Comparison Value)

Using Equation 14, we have calculated the relative fuzzy weights of each criterion. Equation 14 [49] provides the values of the relative fuzzy weights of the identified criteria.

$$\bar{w}_i = \bar{r}_i * (\bar{r}_1 + \bar{r}_2 + \dots + \bar{r}_n)^{-1} \quad (14)$$

The values of the relative weights of the fuzzy comparison matrix are shown in Table 7.

Table 7. Relative fuzzy weights of the criteria/factors

| Criteria   | Wi    |       |       |
|------------|-------|-------|-------|
| Cost       | 0.355 | 0.433 | 0.515 |
| Time       | 0.339 | 0.413 | 0.507 |
| Preference | 0.043 | 0.055 | 0.079 |
| Resource   | 0.068 | 0.099 | 0.141 |

#### 5.2.1.5. Step-5 (Average and Normalized Weights of Criteria)

Using Equation 15 [49], we have calculated the average and normalized weights of the criteria.

$$M_1 = \frac{w_1 + w_2 + \dots + w_n}{\dots}$$

The average and normalized weights of the criteria are shown in Table 8.

Table 8. Average normalized weights of the criteria/factors

| Class      | Mi    | Ni    |
|------------|-------|-------|
| Cost       | 0.434 | 0.428 |
| Resource   | 0.420 | 0.413 |
| Time       | 0.059 | 0.058 |
| Preference | 0.103 | 0.101 |

#### 5.2.1.6. Step-6 (Determine the Local and Global Weights)

Local weights of the criteria determine the weights of the alternatives according to one single criterion. Conversely, the global weights are the aggregated weights and ranking of the alternatives, i.e., crowd-based requirements. The local weights of all the requirements and their average and normalized values are calculated using the same procedure as the one for the factors. The normalized weights of requirements have been calculated individually for each

criterion. Next, we compiled the results obtained from the four-factor ranking. The aggregate weights of the requirements (i.e., the global weights) were calculated by taking the sum product of all alternative weights. After aggregating the weights of the requirements, their total relative score was obtained. Finally, we obtain a prioritized list of requirements according to the four factors namely cost, time, preference, and resource/skill availability.

#### 5.2.1.7. Step-7(Prioritized Requirements)

After computing and arranging (in ascending order) the aggregated results, we obtained a set of prioritized requirements having minimal biases and subjectivity. This is due to the reason that the requirements have been prioritized by a group of expert and novice people to attain the insight of multiple personnel. Table 9 presents the prioritized list of requirements. Evidently, R11 has the highest aggregated priority value with a relative weight of “0.048415009” and R77 has the lowest aggregated score with a relative weight of “0.005648315”.

Table 9. Prioritized list of requirements

|            |      |     |     |     |     |     |     |     |     |     |
|------------|------|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| <b>L1</b>  | R11  | R5  | R14 | R4  | R13 | R21 | R23 | R47 | R28 | R50 |
| <b>L2</b>  | R40  | R33 | R43 | R25 | R34 | R3  | R22 | R46 | R79 | R80 |
| <b>L3</b>  | R44  | R45 | R12 | R1  | R29 | R66 | R37 | R24 | R59 | R81 |
| <b>L4</b>  | R19  | R32 | R38 | R92 | R20 | R42 | R9  | R10 | R48 | R49 |
| <b>L5</b>  | R57  | R2  | R15 | R41 | R58 | R8  | R65 | R56 | R60 | R82 |
| <b>L6</b>  | R83  | R63 | R39 | R26 | R30 | R84 | R36 | R55 | R61 | R62 |
| <b>L7</b>  | R64  | R93 | R6  | R88 | R73 | R17 | R27 | R85 | R87 | R72 |
| <b>L8</b>  | R89  | R77 | R95 | R90 | R7  | R67 | R99 | R98 | R86 | R51 |
| <b>L9</b>  | R18  | R31 | R75 | R68 | R74 | R52 | R53 | R54 | R70 | R71 |
| <b>L10</b> | R100 | R78 | R94 | R91 | R16 | R69 | R96 | R97 | R35 | R76 |

## 6. DISCUSSION

The main aim of this work is to prioritize Crowd-Based Requirements subjectively. To accomplish this, we adopt an effective requirement prioritization technique (F-AHP) by considering multiple perspectives, which supports reducing human bias during the requirements prioritization process. Moreover, F-AHP ranks the Crowd-based requirements grounded on the identified factors from the SLR. Each identified factor has a specific weight in the Crowd-based requirements prioritization process.

For example, the relative weight of *cost* is calculated as “0.427515534” and that of *resource availability* is “0.101242635”. Conversely, the relative weight of *time* and *preference* is “0.413110958” and “0.058130876”, respectively. All the requirements are then mapped against each of the factors respectively and each of their relative weights is compiled and then ranked accordingly.

### 6.1. Identification of Factors

A SLR was conducted to identify the influential factors/criteria that affect the requirements prioritization process. The factors were reported by researchers in the context of CBRE. The most frequently mentioned factors with the highest impact were identified from the literature, which acts as a sole criterion for prioritization of Crowd-based requirements using the proposed quantitative framework. The four identified factors, namely cost, time, resource, and preference, have been further validated by the practitioners and have attained their relative weights according to their level of importance.

### 6.2. Level of Impact of Factors on Prioritization Process

A questionnaire was distributed amongst relevant personnel to identify the level of impact on the requirements prioritization process. The results describe the percentage of the impact of the factors/criteria on the requirements during the prioritization process.

### 6.3. Prioritization of Requirements

After the application of the F-AHP technique, and multi-criteria and multi-perspective prioritization process, the obtained data of each requirement against each criterion increases the maturity and authenticity of the obtained results. After the application of the F-AHP process, it was observed that R11 achieved the highest aggregated relative weight of “0.048415009” since it is documented as the highest preference according to the practitioners. In contrast, R76 has a relative value of “0.001290066” reported to be the lowest value among the considered requirements (Table 9).

### 6.4. Comparison of Proposed F-AHP Technique with AHP

After the implementation of the proposed F-AHP technique, it was observed that both approaches yielded similar results; however, the main difference is in the data collection phase. AHP requires crisp values, whereas F-AHP can be implemented in case of uncertainties. As the CBRE domain contains uncertainties due to the involvement of the crowd; therefore, F-AHP seems more applicable. In comparison with AHP, it was observed that the AHP approach causes ambiguities in the prioritization of requirements due to the selection of alternatives by the decision-makers. Fuzzy set theory can capture this uncertainty and vagueness in human thinking. It also emphasizes the need for a comprehensive framework for structuring a decision problem and quantifying the alternatives essential in natural language judgments.

## 7. Conclusion and Future Work

Crowdsourcing is swiftly shaping up to be the new trend in the technological world and it is being implemented in all fields. In the last decade, it has received significant attention from researchers, especially in the requirements engineering domain, which helped in coining the notion of Crowd-Based Requirements Engineering. In the existing literature, the requirements prioritization process is limited to the opinions or preferences of a single person. Thereby, the current state-of-the-art lacks in providing a quantifiable process for requirements prioritization, which involves multiple factors to gain the multiple perspectives of the crowd. Motivated by this, we proposed a quantitative framework to quantify the requirements prioritization process. First of all, critical factors that affect the requirements prioritization process, including cost, time, skill, preference, and dependency, were identified in the data collection process via a systematic literature review. The data set of requirements was obtained from a crowd for the development of a smart home case study. In total, 100 requirements were considered for the prioritization process from a group of crowd representatives.

The prioritization values of each criterion were obtained from people with different roles. After obtaining the requirement values, the proposed framework adopted an F-AHP technique for the requirements against each criterion. Finally, the results were aggregated to obtain a prioritized list of crowd-based requirements using F-AHP and Multiple Perspective Prioritization Technique. This resulted in a list of an unbiased and prioritized list of Crowd-Based requirements grounded on multiple-criteria and multi-perspective prioritization.

In the future, we plan to validate the proposed quantitative framework by conducting a comparative study using other prioritization baseline techniques. Another future direction is the application of the proposed framework in other domains such as Global Software Development or Agile Software Development. Moreover, we plan to consider other Multi-Criteria Decision Making techniques such as Fuzzy Topsis. Furthermore, we intend to apply the proposed framework to a more diverse data set of crowd-based requirements, which is not limited to the role of respondents or domain.

## REFERENCES

- [1] P. Vogel, and C. Grotherr, “Collaborating with the crowd for software requirements engineering: a literature review,” Designing openness-infusing socio-technical artifacts, PhD dissertation, Staats-und Universitätsbibliothek Hamburg Carl von Ossietzky, 2021.
- [2] U. S. Ghanyani, M. Murad, and W. Mahmood, “Crowd-based requirement engineering,” *Int. J. Edu.*

- Mgt. Eng., vol. 3, pp. 43–53, 2018, [Article \(CrossRef Link\)](#)
- [3] A. Rosmadi, S. S. S. Ahmad, and S. A. Asmai, “Crowdsourcing for requirements engineering: a simplified review,” vol. 10, pp. 134–142, 2018, [Article \(CrossRef Link\)](#)
- [4] I. Morales-Ramirez, F. M. Kifetew, and A. Perini, “Speech-acts based analysis for requirements discovery from online discussions,” *Inf. Sys.*, vol. 86, pp. 94–112, 2019, [Article \(CrossRef Link\)](#)
- [5] E. C. Groen, and M. Koch, “How requirements engineering can benefit from crowds,” *Req. Eng. Magazine*, vol. 8, pp. 10, 2016.
- [6] G. D. Saxton, O. Oh, and R. Kishore, “Rules of crowdsourcing: models, issues, and systems of control,” *Inf. Sys. Mgt.*, vol. 30, no. 1, pp. 2–20, 2013, [Article \(CrossRef Link\)](#)
- [7] E. C. Groen, J. Doerr, and S. Adam, “Towards crowd-based requirements engineering a research preview,” presented at the 21<sup>st</sup> Int. Working Conf. on Req. Eng., 2015, pp. 247–253.
- [8] S. A. Saeed, S. U. R. Khan and, A. Mashkoo, “A Fuzzy AHP-based approach for prioritization of cost overhead factors in agile software development,” *Applied Soft Computing*, vol. 133, pp. 1–20, 2023, <https://doi.org/10.1016/j.asoc.2022.109977>
- [9] W. Maalej, and H. Nabil, “Bug report, feature request, or simply praise? on automatically classifying app reviews,” presented at the 23<sup>rd</sup> Int. Requirement Engineering Conf., 2015, pp. 116–125.
- [10] C. Li, L. Huang, J. Ge, B. Luo, and V. Ng, “Automatically classifying user requests in crowdsourcing requirements engineering,” *J. Sys. Soft.*, vol. 38, pp. 108–123, 2018, [Article \(CrossRef Link\)](#)
- [11] S. Taj, Q. Arain, I. Memon, and A. Zubedi, “To apply data mining for classification of crowdsourced software requirements,” presented at the 8<sup>th</sup> Int. Conf. on Software and Inf. Eng. 2019, pp. 42–46.
- [12] J. Zhang, Y. Wang, and T. Xie, “Software feature refinement prioritization based on online user review mining,” *Inf. Soft. Tech.*, vol. 108, pp. 30–34, 2019, [Article \(CrossRef Link\)](#)
- [13] D. Munante, A. Siena, F. M. Kifetew, A. Susi, M. Stade, and N. Seyff, “Gathering requirements for software configuration from the crowd,” presented at the 25<sup>th</sup> Int. Req. Eng. Conf. Workshop, 2017, pp. 176–181.
- [14] J. Fernandes, D. Duarte, C. Ribeiro, C. Farinha, J. M. Pereira, and M. M. da Silva, “A game-based approach towards improving collaboration and participation in requirement elicitation,” *Proc. Comp. Sc.*, vol. 15, pp. 66–77, 2012, [Article \(CrossRef Link\)](#)
- [15] R. Sharma, and A. Sureka, “Cruise: a platform for crowdsourcing requirements elicitation,” presented at the 10<sup>th</sup> Int. Conf. Contemporary Comp., 2017, pp. 1–7.
- [16] W. Peng, T. Sun, S. Revankar, and T. Li, “Mining the “voice of the customer” for business prioritization,” *ACM Trans. Int. Sys. Tech.*, vol. 3, no. 2, pp. 1–17, 2012, [Article \(CrossRef Link\)](#)
- [17] C. Musto, G. Semeraro, P. Lops, and M. de Gemmis, “Crowdpulse: a framework for real-time semantic analysis of social streams,” *Inf. Sys.*, vol. 54, pp. 127–146, 2015, [Article \(CrossRef Link\)](#)
- [18] N. Makrynioti, A. Grivas, C. Sardianos, N. Tsirakis, I. Varlamis, V. Vassalos, V. Pouloupoulos, and P. Tsantilas, “Palopro: a platform for knowledge extraction from big social data and the news,” *Int. J. Big Data Int.*, vol. 4, no. 1, pp. 3–22, 2017, [Article \(CrossRef Link\)](#)
- [19] M. Z. Kolpondinos, and M. Glinz, “Garuso: a gamification approach for involving stakeholders outside organizational reach in requirements engineering,” *Req. Eng.* vol. 25, no.2, pp. 185–212, 2020, [Article \(CrossRef Link\)](#)
- [20] I. Ibriwesh, S.-B. Ho, I. Chai, and C.-H. Tan, “Prioritizing solution-oriented software requirements using the multiple perspective prioritization technique algorithm: an empirical investigation,” *Conc. Eng.*, vol. 27, no. 1, pp. 68–79, 2019, [Article \(CrossRef Link\)](#)
- [21] M. Dabbagh, S. P. Lee, and R. M. Parizi, “Functional and non-functional requirements prioritization: an empirical evaluation of ipa, ahp-based, and ham-based approaches,” *Soft comp.* vol. 20, no. 11, pp. 4497–4520, 2019, [Article \(CrossRef Link\)](#)
- [22] D. Mougouei, D. M. Powers, and E. Mougouei, “A fuzzy framework for prioritization and partial selection of security requirements in software projects,” *J. Int. F. Sys.*, vol. 37, no. 2, pp. 2671–2686, 2019, [Online] Available: <http://doi.org/doi:10.3233/JIFS-182907>
- [23] A. A. Khan, M. Shameem, R. R. Kumar, S. Hussain, and X. Yan, “Fuzzy ahp based prioritization and taxonomy of software process improvement success factors in global software development,” *App. S. Comp.*, vol. 83, 105648, 2019, [Article \(CrossRef Link\)](#)
- [24] B. Kitchenham, O. P. Brereton, D. Budgen, M. Turner, J. Bailey, and S. Linkman, “Systematic literature reviews in software engineering—a systematic literature review,” *Inf. Soft. Tech.*, vol. 51, no. 1, pp. 7–15, 2009, [Article \(CrossRef Link\)](#)
- [25] G. M. Kanchev, P. K. Murukannaiah, A. K. Chopra, and P. Sawyer, “Canary: extracting requirements-related information from online discussions,” presented at the 25<sup>th</sup> Int. Req. Eng. Conf., 2017, pp. 31–40.
- [26] P. Busetta, F. M. Kifetew, D. Munante, A. Perini, A. Siena, and A. Susi, “Tool-supported collaborative requirements prioritization,” presented at the 41<sup>st</sup> Annual Comp. Soft. and App. Conf., 2017, pp. 180–189.
- [27] J. A. Khan, L. Liu, L. Wen, and R. Ali, “Conceptualizing, extracting and analyzing

- requirements arguments in users' forums: the crowd re framework," J. Soft. Evo. Proc., vol. 32, no. 12, e2309, 2020, [Article \(CrossRef Link\)](#)
- [28] M. Levy, I. Hadar, and D. Te'eni, "A gradual approach to crowd-based requirements engineering: The case of conference online social networks," presented at the 1<sup>st</sup> Workshop on Crowd-based Req. Eng., 2015, pp. 25–30.
- [29] L. Machado, R. Prikladnicki, F. Meneguzzi, C. R. de Souza, and E. Carmel, "Task allocation for crowdsourcing using ai planning," presented at the 3<sup>rd</sup> Int. Workshop on Crowd-Sourcing in Soft. Eng., 2016, pp. 36–40.
- [30] S. Hassan, S. Awan, F. Jaan, and H. Akmal, "Factors affect on requirement prioritization," J. Eng., vol. 2, no. 12, pp. 28–33, [Online] Available: [https://www.iosrjen.org/Papers/vol2\\_issue12\(part-2\)/E021222833.pdf](https://www.iosrjen.org/Papers/vol2_issue12(part-2)/E021222833.pdf)
- [31] H. Zhang, M. Zhang, T. Yue, S. Ali, and Y. Li, "Uncertainty- wise requirements prioritization with search," ACM Trans. Soft. Eng. Meth., vol. 30, no. 1, pp. 1–54, 2020, [Article \(CrossRef Link\)](#)
- [32] J. Linaker, and K. Wnuk, "Requirements analysis and management for benefiting openness," presented at the 24<sup>th</sup> Int. Req. Eng. Conf. Workshops, 2016, pp. 344–349.
- [33] N. Ali, and J.-E. Hong, "A bird's eye view on social network sites and requirements engineering," ICISOFT, 2019, pp. 347–354.
- [34] W. Shim, "An agile method of representing, organizing, and (re) prioritizing requirements in a large enterprise," presented at the 27<sup>th</sup> Int. Req. Eng. Conf., 2019, pp. 464–469.
- [35] S. Scalabrino, G. Bavota, B. Russo, M. Di. Penta, and R. Oliveto, "Listening to the crowd for the release planning of mobile apps, IEEE Trans. Soft. Eng., vol. 45, no. 1, pp. 68–86, 2017, [Article \(CrossRef Link\)](#)
- [36] E. Guzman, M. Ibrahim, and M. Glinz, "Prioritizing user feedback from twitter: a survey report," presented at the 4<sup>th</sup> Int. Workshop on Crowd Sourcing in Soft. Eng., 2017, pp. 21–24.
- [37] P. K. Murukannaiah, N. Ajmeri, and M. P. Singh, "Toward automating crowd re," presented at the 25<sup>th</sup> Int. Req. Eng. Conf., 2017, pp. 512–515.
- [38] V. Gupta, J. M. Fernandez-Crehuet, T. Hanne, and R. Telesko, "Fostering product innovations in software startups through freelancer-supported requirement engineering," R. Eng., vol. 8, 100175, 2020, [Article \(CrossRef Link\)](#)
- [39] R. Snijders, F. Dalpiaz, S. Brinkkemper, M. Hosseini, R. Ali, and A. Ozum, "Refine: a gamified platform for participatory requirements engineering," presented at the 1<sup>st</sup> Workshop on Crowd-Based Req. Eng., 2015, pp. 1–6.
- [40] J. Dabrowski, F. M. Kifetew, D. Munante, E. Letier, A. Siena, and A. Susi, "Discovering requirements through goal-driven process mining," presented at the 25<sup>th</sup> Int. Req. Eng. Conf. Workshops 2017, pp. 199–203.
- [41] A. C. Kumari, and K. Srinivas, "Comparing the performance of quantum-inspired evolutionary algorithms for the solution of software requirements selection problem," Inf. Soft. Tech., vol. 76, pp. 31–64, 2016, [Article \(CrossRef Link\)](#)
- [42] F. A. Bukhsh, Z. A. Bukhsh, and M. Daneva, "A systematic literature review on requirement prioritization techniques and their empirical evaluation," Comp. Stand. & Int., vol. 69, pp. 103389, 2020, [Article \(CrossRef Link\)](#)
- [43] J. A. Khan, Y. Xie, L. Liu, L. Wen, "Analysis of requirements-related arguments in user forums," presented at the 27<sup>th</sup> Int. Req. Eng. Conf., 2019, pp. 63–74.
- [44] E. Sodagari, M. Keyvanpour, "Challenges classification of software requirements interaction management using search-based methods," presented at the 5<sup>th</sup> Int. Conf. on Web Research, 2019, pp. 246–251.
- [45] F. M. Kifetew, D. Munante, A. Perini, A. Susi, A. Siena, P. Busetta, and D. Valerio, "Gamifying collaborative prioritization: Does pontification work?" presented at the 25<sup>th</sup> Int. Req. Eng. Conf., 2017, pp. 322–331.
- [46] X. Li, Z. Zhang, and K. Stefani, "Mobile app evolution analysis based on user reviews," New Trends in Intelligent Software Methodologies, Tools and Techniques, vol. 303, IOS Press, 2018, [Online] Available: <http://doi.org/10.3233/978-1-61499-900-3-773>
- [47] R. Chen, Q. Wang, and W. Xu, "Mining user requirements to facilitate mobile app quality upgrades with big data," E. Com. R. App., vol. 38, 100889, 2019, [Article \(CrossRef Link\)](#)
- [48] H. Malik, E. and M. Shakshuki, "Mining collective opinions for comparison of mobile apps," Proc. Comp. Sc., vol. 94, pp. 168–175, 2016, [Article \(CrossRef Link\)](#)
- [49] L. A. Zadeh, G. J. Klir, and B. Yuan, "Fuzzy sets, fuzzy logic, and fuzzy systems," W. Sc., vol. 6, 1996, [Online] Available: <https://doi.org/10.1142/2895>